Dear Students,

Here's a list of some things I have in mind as I prepare the upcoming exam. Keywords and concepts and so on.

I hope it helps!

Best,
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# Exam 1

- Definition of *statistics*, the field of study
- What is a population? A variable? Data? Sample?
- Different kinds of variables. Quantivative vs. categorical.
- How do we communicate descriptions of data? Averages, dot plots, percentiles, etc.
- Parameters and statistics. What are these, and how are they related?
- Location, spread, shape. What do these terms refer to?
- Commonly used notations: $x_1, x_2, \ldots, x_n$ for a data set from a sample of size $n$, $\bar{x}$ for mean. Can you write the mathematical formula for $\bar{x}$ using these notations?
- What about formula for range?
- What are quartiles? Five-number summary?
- Sample variance, sample standard deviation. What are they? Are you comfortable computing them by hand for small data sets?
- Population variance ($\sigma^2$) and population standard deviation ($\sigma$). What's the difference between these parameters and the statistics?
- How to study the relationship between two quantitative variables.
  - What kind of data does one gather?
  - How does one depict those data?
  - How does one describe what one sees?
- Regression analysis, regression equation, regression line. What are these?
- Least squares method. What's going on here?

- The correlation coefficient, $r$, and its square, $r^2$. What do they tell us? How do you compute them? Recall the interpretation of $r^2$ as the proportion of variation in $y$ that is explained by $x$. What does that mean?
- General advice: This list has already included a few things that, in actual practice, one would compute with the help of technology. (Mean, standard deviation, least squares, correlation coefficient, etc.) On this exam, I am more interested in whether you understand what these things actually are—their mathematical formulas, and what they tell us about data. I might ask you to compute them for small examples.
- How to study the relationship between two categorical variables.
  - What kind of data does one gather?
  - How does one depict that data?
  - How does one describe what one sees?
- Sampling. What is meant by "representative sample"? What is a simple random sample? Stratified sample? Clustered?
- Are you comfortable with the basics of probability theory? How do you interpret the statement "an event has probability $p$ of occurring"?
- Know terminology: sample space, simple event, compount event, event, complementary event, mutually exclusive (or disjoint) events, independent events, conditional probability, etc. I may ask you to define some of these terms, but even if I don't, I will certainly *use* them with the expectation that you know what they mean.
- Are you comfortable with notational conventions? For example, the use of letters, like A, to denote events? Can you translate a mathematical expression into a spoken sentence? For example, how would you translate

  $$\mathrm{P}(\mathrm{A}) = 0.2$$

  into English? ("The probability of the event A is 0.2.") Can you translate

  $$\mathrm{P}(\mathrm{A} \cap \mathrm{B}) = 1/3, \qquad \mathrm{P}(\mathrm{A} \cup \mathrm{B}) = 1/7, \qquad \mathrm{P}(\mathrm{A} \cap \mathrm{B}) = \mathrm{P}(\mathrm{A})\mathrm{P}(\mathrm{B}),$$

  and other similar notational expressions into English?
- Can you translate natural English sentences into mathematical expressions such as the ones above?
- Know basic rules of probability:
  - $0 \leqslant \mathrm{P}(\mathrm{A}) \leqslant 1$ for any event A.
  - Suppose A and B are disjoint events. What is $\mathrm{P}(\mathrm{A} \cap \mathrm{B})$ in terms of the probabilities of A and B? What is $\mathrm{P}(\mathrm{A} \cup \mathrm{B})$?

  - Suppose A and B are independent events. What is $P(A \cap B)$? What is $P(A \mid B)$?
  - Suppose A and B are events. What is $P(A \cup B)$ and how does this relate to the answer when A and B are mutually exclusive?
  - Bayes' Rule. What is it? What's it for?
  - ...and so on.

  There are several of these. Are you comfortable with them?

- I suggest you practice "tree diagrams" and "hypothetical tables." These devices make it easier to understand certain unintuitive results from probability theory.